Decision Models for Binary Financial Document Triage: Adaptation, Missed Documents, and Processing Cost

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Abstract

Financial document triage must trade less analyst reading against missed information and processing cost. We compare seven decision-model, finance-encoder and lexical systems under a binary source-only teacher convention. The SEC-dominated study uses 25,000 training documents, separate 1,000/500 selection/calibration cohorts and a revised 2,000-document evaluation. Trainable core systems share 4,000 documents. At 4k, Laya LoRA achieves macro F1 0.7843 versus 0.7645 for word TF-IDF/logistic. The paired F1 difference interval $[-0.0022, 0.0423]$ does not resolve an F1 difference. They miss 268 and 188 of 1543 review-worthy references, while skipping 32.4% and 24.6% of all documents. Laya's warm mean latency is 64.9 times the lexical classifier's on the recorded machine. Larger training subsets separately measure sample efficiency, and calibration-selected skip policies expose workload/miss tradeoffs. The test reuses previously inspected documents; these results measure teacher agreement, not independent confirmation or analyst-validated materiality. We provide inspectable labels, predictions and independent CPU replay.

1 Introduction

Financial document triage assigns incoming information to automatic discard or analyst review. The useful outcome is less reading with few missed developments, at a processing cost compatible with the incoming volume. A model that sends everything to review avoids false negatives but removes no work. Conversely, fast classification is unhelpful if it discards information the analyst should see.

We study a binary, source-content-only review convention across public financial filings, corporate releases, policy announcements and a smaller news sample. The two labels are *routine* and *review-worthy*. An unresolved, potentially significant financial development goes to review; uncertainty is metadata rather than a third routing destination. Historical company context, portfolio exposures and subsequent returns are unavailable to the classifier. The reference labels therefore define a review convention, rather than establish investment materiality.

Our main question is whether adaptation of a pretrained decision model improves this task relative to a finance encoder and a lexical classifier, and whether any improvement warrants its measured processing cost. Seven systems form the core comparison: native Laya, Laya with head adaptation, Laya with low-rank adaptation (LoRA), frozen FinBERT with logistic regression, FinBERT with LoRA, word TF-IDF with logistic regression, and a training-majority reference. Trained core systems share the same 4,000 documents. Larger training subsets address sample efficiency separately from this equal-data comparison.

The contribution is an incremental financial-domain benchmark with explicit reference provenance, event-group controls, adaptation controls and inspectable workload/resource measurements. The study follows an earlier three-class exploration and reuses its test documents. All binary model and policy choices are frozen before the revised evaluation, but this is not a fresh independent confirmation of the earlier findings. We report that boundary alongside the teacher-agreement nature of the task.

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2 Related work

Financial relevance filtering is established. Ahbali et al. [7] use a TF-IDF/SVM relevance filter before corporate credit-event analysis. Del Corro and Hoffart [8] derive financial-news relevance from stock-prediction attention. FinLin [9] provides multi-source, expert-annotated entity relevance and sentiment. These targets differ from whole-document source-only review triage.

Closely related decision-model studies also exist. Nandakishore [10] evaluates Laya Typed-Decisions, teacher agreement, calibration and selective escalation on synthetic workflows. Stuani's engineering benchmark [11] compares Laya, lexical classifiers, encoders and language models on financial complaint routing and fraud flags. Borchers [12] compares adapted decision models with frozen representations and learning curves on research-paper categories. Our task and corpus differ; we do not claim to introduce financial filtering, decision-model comparison or calibration analysis.

Laya's English root checkpoint uses ModernBERT-large and a learned typed-decision head [2, 3]. FinBERT's adaptation to financial text and sentiment [1] makes it a relevant industry comparator, but its sentiment head does not supply review labels. We use its encoder with a task classifier. LoRA [4] and temperature scaling [5] are established methods. Automated annotation still requires task-specific validation [6]; success on another annotation task does not validate this rubric.

3 Task and data

The classifier receives the extracted English source body. Clearly ordinary or administrative content is routine. A potentially significant financial development, including an unresolved one, warrants review. A scheduled release can carry important information; positive or negative sentiment alone does not determine review. Neither issuer-specific history nor source identifiers are additional model features. Source and event metadata support acquisition, grouping and evaluation.

The corpus draws from SEC filings and release exhibits, Federal Reserve releases, ECB institutional/statistical releases, GOV.UK announcements and Wikinews. It contains 25,000 training documents dated through March 2026, 1,000 selection and 500 calibration documents from April–June, and 2,000 test documents from July–September. Related-document groups do not cross these cohorts. Source coverage follows eligible available documents; there is no ECB quota. The added ECB collection supplies 534 English institutional/statistical releases, with author-named and third-party content excluded. Publication/body snapshots are retained with hashes and attribution. For 350 retained training filings, edgartools supplied the captured text; their public links identify verified parent SEC submissions.

Existing definite references are retained. Luna Flex freshly decides every previously unclear reference and labels newly selected documents using a pinned binary rubric. Thus *unclear* is not mapped mechanically into either class. A blinded, balanced 200-document training-only consistency audit compares the retained references with binary decisions; it does not override them. This is a hybrid annotation protocol, and the audit measures teacher consistency rather than agreement with financial analysts. Teacher-reported uncertainty is neither a ground-truth error probability nor an evaluation class.

Evaluation additions are chosen without labels or model scores, retaining the earlier selection/calibration/test memberships. The revised test keeps the previous 1,800 bodies and adds 200. Group checks exclude exact matches, known event aliases and verified near-duplicates; probabilistic candidate retrieval and inherited groups can miss relationships. Unselected raw corpora are not exhaustively re-clustered. SEC content remains dominant, and the small news component does not represent a general news inbox.

Table 1: Frozen cohort support. The test retains 1,800 earlier bodies; reference labels are revised.

Cohort	Documents	Routine	Review	Groups	SEC	ECB
Train	25000	7173	17827	24696	20740	1175
Selection	1000	283	717	993	889	29
Calibration	500	135	365	493	461	12
Test	2000	457	1543	1967	1870	40

4 Systems and evaluation

4.1 Fixed comparisons and adaptation

Training subsets of 4,000, 16,000 and 25,000 documents are nested, with stable round-robin family/class sampling. All trainable core systems use the same 4,000 bodies; the majority reference counts the same labels without optimization. Native Laya requires no task training. TF-IDF and frozen FinBERT/logistic additionally use 16,000 and 25,000 training documents. Both LoRA systems additionally use 16,000; their curves stop at that size. Every learning-curve point also uses the same 1,500 development labels; training count is not total annotation cost.

TF-IDF learns its vocabulary and transforms from training alone. Logistic regularization and class weighting use the same bounded grid at each training size, selected by selection macro F1 (Appendix A). Frozen FinBERT features are computed once and cached with body, recipe and source identities. Feature-preparation durations and cache use are itemized; command time includes encoding and classifier search. Frozen encoder windows are averaged and L2-normalized before classification.

We use pinned `convaiinnovations/laya` English root and `ProsusAI/finbert` checkpoints; full revisions accompany the evidence. Adapted neural systems use seed 42, unweighted document-level cross-entropy and up to three epochs. Raw selection macro F1 chooses the checkpoint; epoch zero is eligible, and ties retain the earlier epoch. Laya head adaptation updates the decision head with a frozen encoder at learning rate 3×10^{-5}. LoRA updates rank-eight encoder adapters and the task head at 10^{-4}, with scaling 16 and adapter dropout 0.05. Matched pre-update states are preserved as supporting controls, avoiding a direct attribution of every SDK-to-adapted change to weight updates.

Laya adaptation uses a 1,024-token context and FinBERT uses 512. Overlapping windows cover all normalized source tokens, and uniform window-logit pooling gives one document-level prediction and loss. No window labels are inferred. The native Laya SDK uses 512-token windows and limits GPU passes to eight windows, retaining its most-confident long-document aggregation. It therefore differs from adapted Laya in context and pooling as well as fitting. Architectures, parameter counts, pretraining objectives and classifier recipes differ across systems; those comparisons cannot isolate finance pretraining or LoRA as a sole cause.

4.2 Agreement, misses and review workload

Macro F1 averages the routine and review-worthy class F1 values. Exact binary ties route to review. We also report review precision/recall, the 2×2 confusion matrix, and false negatives divided by review-worthy reference support. At the default decision, a document predicted routine is skipped. The skipped fraction quantifies reading removed; its complement is review workload. A low miss rate can reflect more review rather than better discrimination, so both quantities accompany every operational comparison.

Scalar temperature is fit by calibration negative log likelihood. It changes confidence while preserving the default class decision. Binary Brier score is the sum over both class columns, equal to twice the positive-class squared error. We report raw/calibrated scoring arithmetic in the evidence package.

We fit five routine-only skip policies on calibration: empirical review-miss targets of 1% and 5%, and maximum skip budgets of 10%, 20% and 30%. Each chooses the largest feasible calibration skip

count without splitting equal-probability ties. Predicted review documents are never skipped by these rules. We report achieved test skip counts and missed references, rather than assume workloads match or that fitting targets are population guarantees.

The fixed primary comparison is 4k Laya LoRA versus 4k word TF-IDF/logistic. We use 2,000 paired whole-group bootstrap draws for percentile 95% intervals and their differences. These intervals condition on a single fitted run; they do not measure training instability or correct supplementary comparisons for multiplicity. Source/length slices are descriptive, with support reported. No method, checkpoint or policy is selected using revised-cohort outcomes.

4.3 Measured resources

GPU jobs run serially on an NVIDIA RTX 3060 with 12 GiB GPU memory; CPU work uses an Intel Core i5-12400F. Warm inference includes tokenization/windowing and classification, excludes model loading, and uses the same 128 selection bodies. We retain each latency, mean/P95, batch-throughput denominator and peak allocated GPU memory. Recorded fit-phase and successful command elapsed times are distinguished; interrupted attempts are accounted for separately. Frozen feature preparation/cache measurements accompany classifier fitting. Local timings on the recorded hardware do not estimate cloud prices, energy use or deployment service latency.

5 Results

5.1 Equal-data system comparison

Laya LoRA has the highest 4k macro F1 point estimate in Table 2. The prespecified Laya-minus-TF-IDF contrast is 0.0199 with paired 95% interval $[-0.0022, 0.0423]$. The interval includes zero and does not establish equivalence. At default decisions, Laya skips 156 additional documents but misses 80 more review-worthy references. Its review precision is 94.3% versus 89.9%, while recall is 82.6% versus 87.8%. The higher macro F1 therefore does not imply fewer missed developments. Laya's warm mean inference latency is 64.9 times the lexical classifier's.

Table 2: Core systems on the revised 2,000-document evaluation. Default decisions; false negatives (FN) are divided by review-reference support. All trainable systems use the same 4k bodies; majority counts those labels, and SDK has no task fit. Warm inference includes full document processing.

System	F1 (95% CI)	FN	Skipped	Mean ms	P95 ms
Training majority	0.436 [0.429, 0.441]	0/1543	0.0%	2.97	3.45
Word TF-IDF/logistic	0.764 [0.743, 0.786]	188/1543	24.6%	4.58	8.16
Frozen FinBERT/logistic	0.718 [0.696, 0.740]	302/1543	30.9%	35.81	113.33
FinBERT LoRA	0.743 [0.721, 0.763]	352/1543	36.5%	35.17	116.72
Laya native SDK	0.542 [0.519, 0.567]	96/1543	8.3%	334.22	1332.02
Laya head	0.609 [0.587, 0.632]	473/1543	37.3%	276.66	1131.21
Laya LoRA	0.784 [0.764, 0.805]	268/1543	32.4%	297.19	1207.55

5.2 Review workload and missed references

Table 3 applies calibration-selected thresholds unchanged to test. The 1% fitting target and 20% skip budget produce different achieved workloads across systems; neither is a test guarantee. For the primary pair, the 1% calibration miss target reduces skipped test reading to 2.5% for Laya and 6.2% for TF-IDF. Stringent miss control removes much of the workload reduction apparent at default decisions. A lower achieved miss count cannot be interpreted independently of how much reading remains.

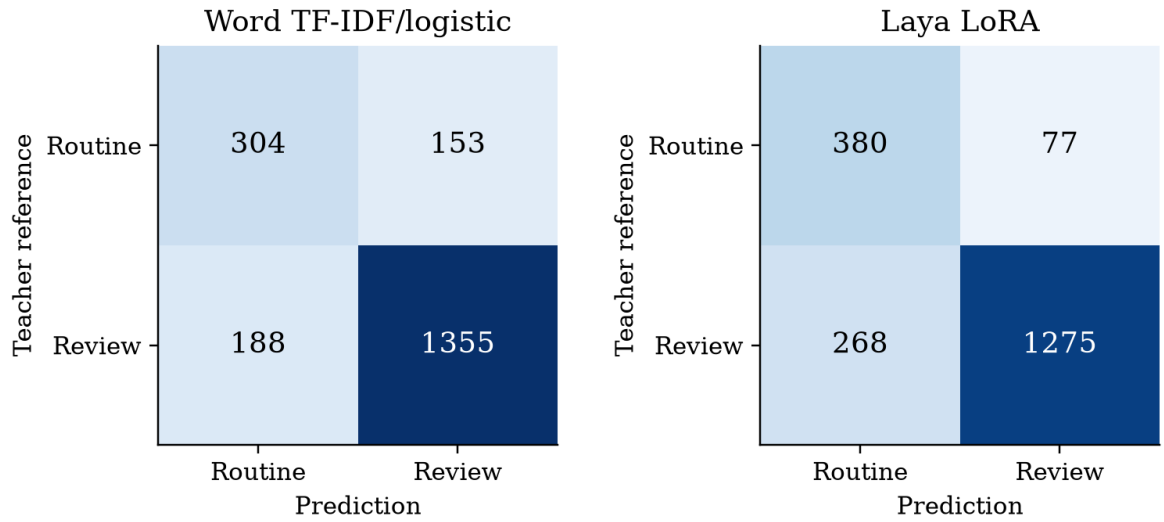


Figure 1: Default two-class confusion matrices for the fixed primary pair. Rows are teacher references; columns are model predictions. Counts sum to 2,000 in each matrix.

Table 3: Achieved test outcomes for two calibration policies. The left pair uses an empirical 1% calibration miss target; the right pair maximizes calibration skips within 20%. All policies skip only routine predictions and preserve ties. Achieved test workloads are not assumed equal.

System	1% miss: FN	Skipped	20% skip: FN	Skipped
Training majority	0/1543	0.0%	0/1543	0.0%
Word TF-IDF/logistic	20/1543	6.2%	95/1543	15.8%
Frozen FinBERT/logistic	27/1543	4.0%	159/1543	18.7%
FinBERT LoRA	31/1543	5.3%	144/1543	19.8%
Laya native SDK	14/1543	0.9%	95/1543	8.2%
Laya head	14/1543	1.4%	240/1543	19.6%
Laya LoRA	7/1543	2.5%	124/1543	19.7%

5.3 Adaptation and sample efficiency

Figure 2 separates training size from the shared-4k comparison. Intervals are conditional on fixed fitted runs and are not corrected for multiple comparisons. Training source/class composition and optimization steps also change across sizes; these are fixed-recipe curves, rather than a causal estimate of sample count alone.

At 16k, Laya LoRA reaches F1 0.8424 and word TF-IDF/logistic 0.8245; FinBERT LoRA reaches 0.8140. Laya and TF-IDF default miss counts are 171 and 141, with 28.0% and 24.3% skipped. The successful commands take 20.66 hours and 59.5 seconds, respectively; FinBERT LoRA takes 2.51 hours. These larger-run point estimates supplement the fixed 4k primary comparison; they do not replace it.

The matched pre-update comparisons in Table 5 separate task fitting from context, pooling and initial-head differences. Each recipe is compared with its own preserved initial state. These controls do not isolate pretraining or equalize classifier budgets across model families.

The primary systems share 122 default missed references, with 146 additional Laya-only and 66 TF-IDF-only misses. This overlap describes the fixed predictions; it does not evaluate a new combined routing rule.

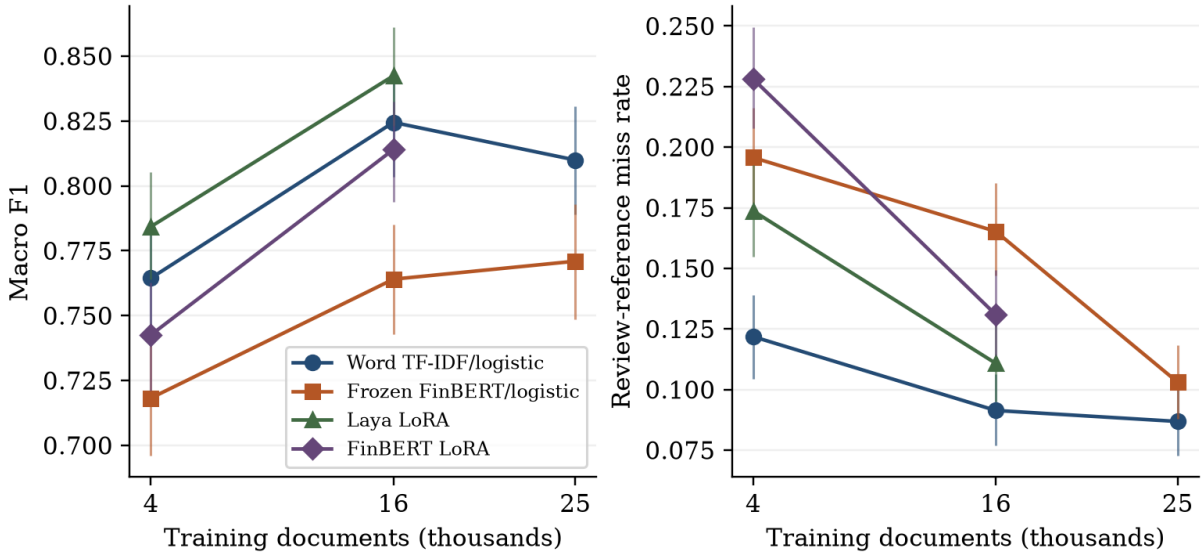


Figure 2: Nested training-size curves for completed recipes. Left: macro F1. Right: review-reference miss rate at default decisions. Vertical segments are marginal whole-group 95% intervals. LoRA curves stop at 16,000 training documents.

Table 4: Paired training-size F1 increments for fixed system families. Conditional 95% intervals; supplementary comparisons are uncorrected.

System	Training change	Δ F1	95% interval
Word TF-IDF/logistic	4k to 16k	0.0600	[0.0421, 0.0795]
Word TF-IDF/logistic	16k to 25k	-0.0145	[−0.0295, 0.0013]
Frozen FinBERT/logistic	4k to 16k	0.0458	[0.0292, 0.0624]
Frozen FinBERT/logistic	16k to 25k	0.0069	[−0.0098, 0.0230]
Laya LoRA	4k to 16k	0.0581	[0.0428, 0.0735]
FinBERT LoRA	4k to 16k	0.0715	[0.0535, 0.0899]

6 Discussion and limitations

The lexical classifier remains a credible low-cost reference on this benchmark. Laya adaptation improves agreement relative to its matched initial state, but greater reading reduction at default decisions also brings additional misses. A triage operating point therefore needs an explicit miss/workload objective fixed before evaluation; a macro F1 ranking alone cannot determine it. The training-only reference audit agrees on 93.0% of 200 balanced retained labels. That consistency result cannot substitute for expert reference validity or estimate a population annotation error rate. The reference convention is the central limit. Training and evaluation share a teacher, and retained versus newly decided labels use different protocols. Independent expert adjudication is needed before treating a teacher miss as an economically material miss. Review priorities also depend on company, portfolio and historical context absent from these inputs. No trading strategy or investment profitability is evaluated.

The revised evaluation is temporally later but reuses previously inspected test bodies. Fixing binary choices before the rerun prevents selection on its new scores; it does not erase earlier information or create independent confirmation. Fresh representative evaluation data would be needed for that claim. Source imbalance, public-document selection, revised historical web content, possible pretraining exposure and missed duplicate relationships further constrain transfer to industry inboxes.

The fixed recipes answer a bounded comparison, not the best achievable score for every model family. One fitted seed leaves optimization variability unmeasured. Supplementary scaling/control intervals are uncorrected, and calibration operating points can shift with source mix. The study includes neither a

frontier generative classifier nor full-model adaptation or RLCD training. It should guide a narrower next experiment, rather than establish a universal model ranking.

7 Conclusion

On this SEC-dominated teacher benchmark, LoRA improves Laya relative to matched pre-update states and benefits from the larger training subset. The prespecified 4k comparison does not establish an F1 advantage over word TF-IDF/logistic; Laya's additional default skipping comes with more missed references and substantially higher measured processing cost. Calibration exposes how stringent miss control reduces reading removed. Expert reference validation and a fresh representative evaluation cohort are needed before operational generalization claims.

Code and reproducibility

The public repository supplies a separate binary evidence package with ordered references/logits, decisions, fitting histories, resource measurements, artifact/source identities and independent CPU arithmetic replay. Exact model fitting additionally requires the preserved local bodies, checkpoints and pinned pretrained weights. Public source URLs identify the documents, but live recollection does not guarantee identical historical text. Attribution and transfer conditions accompany collection code; bodies and weights are excluded from the public evidence package. The earlier three-class study and its scientific bundles remain preserved. The command `python decision_models/reports/binary-v2/verify.py` replays the published metrics, policies, timing summaries and paired intervals on CPU without API calls. Code and evidence are available in the versioned study package. The revised freeze identity is `e6afdaf08581`; its full hash and source identities accompany the package.

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A Supporting comparisons and provenance

The following tables support the main agreement/workload questions: recipes, pre-update controls and fitting resources. Full calibration policies, descriptive source/length/reference-basis slices and per-document arithmetic are retained in the public evidence package with their audit receipts. The linear grid uses $C \in \{0.001, 0.01, 0.1, 1, 10, 30, 100\}$ and either no class weighting or balanced weights.

Table 5: Matched pre-update comparisons. The initial adapted pooling/head state is the control; it is distinct from native SDK inference. Larger fixed recipes reuse a matched setup's pre-update control. Intervals are supplementary.

System	Train	Zero F1	Fit F1	Δ F1 interval
Laya head	4k	0.557	0.609	[0.023, 0.083]
Laya LoRA	4k	0.557	0.784	[0.196, 0.259]
FinBERT LoRA	4k	0.187	0.743	[0.535, 0.574]
FinBERT LoRA	16k	0.187	0.814	[0.607, 0.646]
Laya LoRA	16k	0.557	0.842	[0.255, 0.316]

Table 6: Completed neural configurations. All use seed 42 and at most three epochs; epoch zero is eligible. A context limit includes task/model overhead, rather than denote full source coverage. All normalized source tokens are windowed.

System	Train	Context	Trainable	Selected epoch
Laya head	4k	1024	26248193	2
Laya LoRA	4k	1024	27165697	3
FinBERT LoRA	4k	512	297986	3
FinBERT LoRA	16k	512	297986	3
Laya LoRA	16k	1024	27165697	3

Table 7: Recorded resources. Command time includes preparation, fitting and development evaluation. Fit phase is the neural history or linear preparation/search duration, including selection overhead. Fit GiB records neural-fit peaks only; '–' denotes an unavailable measurement. Inference peaks can include resident fitting state. Frozen feature/cache costs are included in command time and itemized in the evidence.

System	Train	Command s	Fit phase s	Fit GiB	Infer GiB
Training majority	Count 4k	6.3	0.2	–	0.00
Word TF-IDF/logistic	4k	22.9	15.9	–	0.00
Frozen FinBERT/logistic	4k	219.1	199.1	–	0.51
Laya native SDK	None	812.9	–	–	2.35
Laya head	4k	9225.8	8541.7	2.37	2.10
Laya LoRA	4k	17014.8	16278.2	3.24	2.12
FinBERT LoRA	4k	2209.4	2099.5	0.78	0.52
Word TF-IDF/logistic	16k	59.5	51.8	–	0.00
Word TF-IDF/logistic	25k	97.2	89.8	–	0.00
Frozen FinBERT/logistic	16k	485.9	465.4	–	0.51
Frozen FinBERT/logistic	25k	481.5	459.2	–	0.51
FinBERT LoRA	16k	9043.6	8951.9	0.79	0.52
Laya LoRA	16k	74374.3	73637.6	3.24	2.12

Laya native SDK also used 886.0 seconds in prior interrupted attempts. These attempts are excluded from the successful-command column; cumulative elapsed time and attempt receipts remain in the evidence package.